

The Complement Rule Is Excluded Middle: Machine-Checked Probability over Intuitionistic Logic

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Abstract

Kolmogorov’s probability is defined within a classical logic framework, whose complement rule becomes equivalent to the law of excluded middle in the limiting case of certainty. Quantum mechanics has shown that this logical substrate is just one of many possible logical modeling choices. We mechanize, in Lean 4 over mathlib, what probability becomes when the event logic is *intuitionistic*: valuations on frames, the algebras of verifiable propositions. Two hinges anchor the development: (1) the complement rule $v(A^c) = 1 - v(A)$ holds for every valuation exactly when the algebra satisfies excluded middle, and (2) Dempster conditioning agrees with Bayesian conditioning at a piece of evidence exactly when that evidence carries zero slack. On finite frames, every valuation is a classical mass function on points. In the general case, valuation splits into atomic and diffuse parts. Its Boolean core carries a totally monotone Dempster–Shafer belief function. We define slack as the mass that escapes point-valued certainty: it is boundary mass in the measure model. On a semidecidable event, forcing it to zero is exactly as hard as solving the halting problem. We further prove a dichotomy settling where the probability monad exists and where it fails, show that the naive transcription of Cox’s theorem is unsatisfiable, and build the Aczél/Hölder analytic core from scratch.

Keywords: intuitionistic logic, probability, Cox’s theorem, belief functions, locales, Lean, mathlib, formalized mathematics

1 Introduction

Classical logic underlies Kolmogorov’s probability. The axioms of this probability specify an arithmetic of chance (non-negativity, normalization, additivity). Beneath that arithmetic sits a logic: the events form a Boolean σ -algebra, closed under complement and satisfying excluded middle. The complement rule, $P(A^c) = 1 - P(A)$, is where the two layers touch. It is an arithmetical equation about a logical operation, obeyed by every assignment of degrees of certainty. In the limiting case of all-or-nothing probabilities, it becomes equivalent to the law of excluded middle.

There is no intrinsic reason why probability should be compatible with exactly one logic as different situations are described by different logics. Intuitionistic logic describes computation, verification, and finite-precision observation, where a proposition may be confirmable but not refutable. Linear logic describes resources, where using an assumption

consumes it [Girard 1987]. Quantum logic describes measurement, where propositions fail to distribute. If probability is, as the Cox–Jaynes tradition has it, *extended logic* [Cox 1946; Jaynes 2003], the calculus of degrees of certainty over a space of propositions, then the logic of those propositions is a *parameter* of the theory, and Kolmogorov’s calculus is the value of that parameter at Boolean logic.

Physics has already introduced a use case for a logic not based in Boolean algebra. Quantum probability lives on orthomodular lattices of projections rather than Boolean algebras [Birkhoff and von Neumann 1936]. Gleason’s theorem [Gleason 1957] does for quantum probability what this paper’s representation theorems do here: it identifies the states compatible with the event logic. It seems natural to conclude that the Boolean substrate is a modeling choice. In this work, investigate what happens when we change this parameter. Specifically, we ask “what does probability become when the logic of events is *intuitionistic*?”.

Intuitionistic logic is the natural first non-Boolean instance to study using the proofs-as-programs paradigm. The choice of (in our case, intuitionistic) logic sits upstream of statistics the way a type system sits upstream of programs: the same syntax of operations, a different set of valid transformations, and some derivations that typecheck classically can either fail or split in two. This logic’s algebraic semantics is mature order theory, well supported by mathlib [mathlib Community 2020]: frames, that is, complete Heyting algebras, modelled on the lattices of open sets. The content of the original (classical) probability definitions and results is not lost: complete Boolean algebras are exactly the frames where excluded middle holds, so the classical theory embeds as a special case rather than being replaced.

The choice of logic fixes the event algebra. What remains is to fix the arithmetic on top of it, keeping only the axioms that can be stated without a complement. A *valuation* on a frame Ω is a monotone, normalized, modular map $v : \Omega \rightarrow [0, \infty]$. For v , modularity means inclusion–exclusion, $v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b)$, and no axiom mentions a complement. In a Heyting algebra the pseudo-complement a^c satisfies $a \sqcap a^c = \perp$ but not, in general, $a \sqcup a^c = \top$. The complement rule therefore degrades to a derived *inequality* $v(a) + v(a^c) \leq 1$, with a gap

$$\text{slack}(a) = 1 - v(a) - v(a^c)$$

measuring exactly the valuation of the region where excluded middle fails for a . The term *slack*, and this logical reading, are introduced in this paper. The quantity itself is

not new: it is the width of the classical Dempster–Shafer belief/plausibility interval [Dempster 1967; Shafer 1976]. The two align on the frame’s Boolean core. Dempster–Shafer reads that width as an agent’s ignorance. Here it needs no agent: in the concrete model where Ω is the opens of a space carrying a probability measure μ , the slack of an open U is precisely $\mu(\partial U)$, the measure of its topological boundary. The slack of an open set is a structural quantity, not a measure of ignorance. The theory that results is recognizably the *shape* of Dempster–Shafer belief functions, but derived from the underlying logic rather than postulated.

The statistical *operations* are logic-uniform. The conditional is still $v(a \sqcap b)/v(b)$, and the product rule and Bayes symmetry hold verbatim. What varies is the set of *identities* those operations satisfy, and with it the licensed computations. The classical case split over $\{A, A^c\}$, and with it the usual denominator of Bayes’ theorem $P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$, degrades to a lower bound whose deficit is a conditional slack. Marginalization is exact only over *genuine* partitions, families that actually join to $\top$. The complement trick $P(A) = 1 - P(A^c)$ degrades to the interval $[v(A), 1 - v(A^c)]$ of width $\text{slack}(A)$, so point statistics becomes interval statistics exactly on the events the logic leaves undecided, and collapses back to points on complemented ones. Updating itself forks: Dempster’s rule and Bayesian conditioning, classically indistinguishable, give different posteriors from the same evidence unless its slack vanishes.

The choice of logic changes the query being answered: the value of an event A is $P(\Box A)$, where $\Box A$ is the interior of A , i.e. the probability of its verifiable part. In the measure model, slack is boundary mass, so a nonatomic distribution queried on intervals behaves entirely classically, since an interval’s boundary is two points, which carry no mass. The differences are observable precisely at events with charged boundaries and at semidecidable hypotheses.

A theory this close to excluded middle must be developed with unusual care, because classicality is easy to smuggle in: a single informal case split over $\{A, A^c\}$ silently decides an undecidable event. The risk is documented even classically, as Cox’s original theorem required repair after Halpern’s counterexample [Cox 1946; Halpern 1999; Horn 2003], and the constructive setting multiplies the opportunities. The direct transcription of the classical Cox statement within an intuitionistic logic setting is not merely unprovable but *unsatisfiable*. The sum rule is irreducible: a disjunction’s value is provably not a function of the values of its disjuncts, so modularity must be posited rather than derived. Every appearance of classical logic in this work, including at the meta-theory level, is tracked and justified at the point it is used, as with the computable Decidable instance behind the CoxModel non-vacuity witness of Section 6.

Full excluded middle is not the only stop between plain intuitionistic logic and Boolean logic. Asking what changes at an intermediate stop is a natural companion question

to our main inquiry into intuitionistic probability. Weak excluded middle, $a^{cc} \sqcup a^c = \top$, is one such case. Using our valuation machinery, we can demonstrate that slack collapses to the regularity defect alone under exactly this condition, dropping the independent De Morgan gap.

Contributions. Our contributions are as follows:

1. **The two hinges** (Sections 3 and 5). The complement rule holds for *every* valuation iff excluded middle holds in the frame, an equivalence isolating the exact content of Van Horn’s classical negation axiom R3 [Horn 2003]. Dempster and Bayesian conditioning agree iff the evidence has zero slack;
2. **Structure theorems** (Sections 4 and 5). On finite frames a valuation is a classical mass function on points. In general it splits into atomic and diffuse parts, and on its Boolean core of regular elements it is a belief function;
3. **The Cox program, dissected** (Section 6, Appendix A). The product-rule half is independent of the logic, proved by an Aczél/Hölder generator built from scratch. The sum-rule half (a theorem in classical Cox) admits no such derivation here, as shown by a five-element counterexample, and must therefore be included among the defining axioms of a valuation;
4. **Two grounding models** (Section 7). Any classical probability measure, read on the opens, is a valuation whose slack is boundary mass. On the Sierpiński frame, a slack-free classical readout of halting is uncomputable;
5. **The monad dichotomy** (Section 9). Product valuations are unique on chain products and provably non-unique on the powerset frame, where Fubini fails. The countable fragment composes associatively and commutatively;
6. **Weak excluded middle**. (Sections 2 and 7). The global weak excluded middle condition is exactly extremal disconnectedness, the class of Stonean spaces;
7. **Mechanization**. Everything above is mechanized in Lean 4 using mathlib, and is (to our knowledge) the first mechanization of non-additive probability over intuitionistic logic.

2 Valuations on Frames

2.1 The event logic

A *frame* is a complete lattice in which finite meets distribute over arbitrary joins. Equivalently, it is a complete Heyting algebra. Frames are the algebraic semantics of intuitionistic logic and the point-free abstraction of topological spaces. The same objects with morphisms reversed form the category of *locales*, so the distinction is invisible at the level of a single algebra. We say *frame* throughout and reserve *locale*

for the spatial and categorical statements (points of a locale, classifying locales) where that word is the standard one. The motivating example throughout is $\Omega = \mathcal{O}(X)$, the opens of a space, with $\sqcap, \sqcup$ as intersection and union and the pseudo-complement a^c of an open given by the *interior* of its set complement. In mathlib this is the typeclass `Order.Frame`, and the Heyting operations come with the key asymmetry already in place: $a \sqcap a^c = \perp$ holds in any Heyting algebra (`inf_compl_eq_bot`), while $a \sqcup a^c = \top$ is available only in Boolean algebras. An element with $a \sqcup a^c = \top$ is *complemented*: it satisfies its own instance of excluded middle.

2.2 The valuation structure

The central object assigns a degree of certainty to each element of the frame, with inclusion–exclusion as the only interaction axiom:

Definition 1 (Valuation). A *valuation* on a frame Ω is a function $v : \Omega \rightarrow [0, \infty]$ with $v(\perp) = 0$, $v(\top) = 1$ (monotonicity), and, for all $a, b \in \Omega$, *modularity*:

$$v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b).$$

```
structure Valuation (Ω : Type*) [Order.
  Frame Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono' : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b
    = toFun (a ⊔ b) + toFun (a ⊓ b)
```

Values live in $[0, \infty]$ (`ENNReal`, extended non-negative reals) following mathlib’s measure-theoretic convention, and normalization then confines them to $[0, 1]$ (`Valuation.le_one`). The definition is similar to Kolmogorov’s, with the difference being that *no axiom mentions complement*. Classically, this omission is invisible, because the complement rule is derivable from additivity and $a \vee \neg a = \top$. Constructively, this changes the definition of what probability is.

2.3 Slack, and its two independent obstructions

Modularity applied to the disjoint pair (a, a^c) collapses to $v(a) + v(a^c) = v(a \sqcup a^c)$ (`add_compl_eq_sup`). So, the join need not be $\top$:

Theorem 2 (`add_compl_le_one`). *For every valuation v on a frame Ω and every $a \in \Omega$, $v(a) + v(a^c) \leq 1$.*

This is a *derived* Dempster–Shafer sub-additivity inequality. The deficit

$$\text{slack}(a) = 1 - (v(a) + v(a^c)) = 1 - v(a \sqcup a^c)$$

is the valuation of the undecided region. It decomposes canonically. The pair (a^{cc}, a^c) , double negation against negation, is always disjoint. However, it is a partition only under *weak* excluded middle ($a^c \sqcup a^{cc} = \top$). Applying modularity to this disjoint pair splits the slack into two logically independent gaps:

Theorem 3 (`slack_eq_dnGap_add_deMorganGap`). *For every valuation v on a frame Ω and every $a \in \Omega$, $\text{slack}(a) = \text{dnGap}(a) + \text{deMorganGap}(a)$, where $\text{dnGap}(a) = v(a^{cc}) - v(a)$ is 0 if a is $\neg\neg$ -stable (regular), and $\text{deMorganGap}(a) = 1 - v(a^{cc} \sqcup a^c)$ is 0 if weak excluded middle holds at a .*

The single Dempster–Shafer “ignorance” number is thus resolved into a regularity defect and a De Morgan defect: the first measures failure of double-negation stability, the second corresponds to a named intermediate logic (weak excluded middle). An informal aside on what the first gap measures: constructively, the contrapositive of a universal hypothesis returns only its double-negated form, so for open sets, this specification of probability does not fall victim to Hempel’s raven paradox [Hempel 1945]: evidence of the white sock indoors confirms at most a^{cc} , not a itself, and `dnGap` is exactly the gap between confirming a^{cc} and confirming a itself.

Remark 4 (`Valuation.slack_eq_dnGap_of_isDeMorgan`). The two gaps are logically independent at a single element. However, if weak excluded middle is imposed on the frame for every a ($a^{cc} \sqcup a^c = \top$, the condition `Order.Frame.IsDeMorgan`), then `deMorganGap` vanishes identically and $\text{slack}(a) = \text{dnGap}(a)$ for every valuation and every a . This sits strictly between plain intuitionistic logic, where both gaps can be positive, and full excluded middle, where both vanish (`classical_slack_zero`). We do not build on this regime elsewhere, as it is merely a parallel development and an obvious complement to the fully constructive theory this paper focuses on. It easily follows from decomposition for free, and a concrete instance appears in Section 7.

2.4 What survives without excluded middle

Three classical pillars survive intact, and one fails in a precisely delimited way. *Disjoint additivity* is unconditional: $a \sqcap b = \perp$ implies $v(a \sqcup b) = v(a) + v(b)$. *Conditioning* is native: for $v(b) \neq 0$ the posterior $v(a \mid b) = v(a \sqcap b) / v(b)$ is again a valuation (the proof is frame distributivity), and the product rule $v(a \mid b) \cdot v(b) = v(a \sqcap b)$ and Bayes symmetry (and therefore Bayesian updating) $v(a \mid b) v(b) = v(b \mid a) v(a)$ hold (`condVal_mul`, `condVal_symm`). *Total probability* holds over any genuine partition, meaning $a \sqcap a' = \perp$ and $a \sqcup a' = \top$: $v(b) = v(a \sqcap b) + v(a' \sqcap b)$. What fails is total probability over $\{a, a^c\}$. That family does not tile $\top$, and the conditional masses fall short of $v(b)$ by a conditional slack (`cond_add_compl_le`).

The classical fragment. We characterize the elements of Ω on which the classical complement rule holds. The $\neg\neg$ -stable (*regular*) elements, which form a Boolean algebra (by Glivenko's theorem), are a natural candidate, but they do not suffice. Modularity constrains the frame join $\sqcup$, whereas the Boolean join of regular elements is the regularized $(a \sqcup b)^{cc}$, for which only the inequality $v(a) + v(b) \leq v((a \sqcup b)^{cc}) + v(a \sqcap b)$ survives (i.e. the 2-monotonicity of Section 5). A regular but uncomplemented element, which is an open ray $(-\infty, 0)$ in the opens of $\mathbb{R}$, under an atomic measure at 0, still carries slack. The correct classical fragment is the *complemented* elements, where $v(a) + v(a^c) = 1$ holds exactly (add_compl_eq_one_of_complemented). On a connected locale that fragment can be as small as $\{\perp, \top\}$. In the Boolean limit, every element is complemented, the slack vanishes identically, and Kolmogorov is recovered (classical_additivity, classical_slack_zero).

Mixtures. Valuations are closed under finite convex combination (Valuation.mix), with every axiom inherited termwise, because modularity is linear. This innocuous fact becomes a characterization in Section 4.

3 Where Classicality Lives: the R3 Hinge

Classicality lives at multiple levels in this work: one meta-theoretic, in the proof assistant itself, and one object-level, in the frame's own logic.

Lean's meta-theory. Lean's foundations include Classical.choice, and mathlib is built on it throughout. For example, the order relation on $\mathbb{R}$ and $[0, \infty]$ has no computable Decidable instance in general, so any library lemma that case-splits on $x \leq y$ for reals resolves that split via Classical.propDecidable. A formalization that never invokes excluded middle at the level of frame elements $a \in \Omega$ can therefore still be classical through the arithmetic it borrows from the library. We track this separately from object-level classicality throughout the paper. Theorem 12 of Section 4 rests only on this ambient baseline and contains no explicit case split. The Scott-continuity recovery theorem (Theorem 11) invokes Lean's classical tactic directly, and the CoxModel witness of Section 6 goes the other way, replacing the default with an explicit computable Decidable instance to keep that one instance genuinely non-classical.

Object-level classicality: the R3 axiom. Van Horn's repaired Cox theorem [Horn 2003] axiomatizes plausibility over classical propositions. Its negation axiom, called R3, posits that the plausibility of $\neg A$ is a fixed non-increasing function S of the plausibility of A . Consistency forces the involution $S \circ S = \text{id}$, and therefore the complement rule $p \mapsto 1 - p$. We show that the involutive negation is double-negation elimination, so R3 is where classical logic enters

every known Cox-style derivation. To do this, we first isolate the exact content of R3's conclusion as a predicate on valuations, then prove equivalence to excluded middle.

```
def HasClassicalNegation (v : Valuation Ω)
  : Prop :=
  ∀ a, v a + v ac = 1
```

Theorem 5 (the R3 hinge). *For every frame Ω , every valuation on Ω has classical negation iff $a \sqcup a^c = \top$ for every $a \in \Omega$.*

The forward direction (hasClassicalNegation_of_em) is a calculation. The content is the converse (em_of_forall_hasClassicalNegation): wherever excluded middle fails, a valuation with genuine slack must be *manufactured*. Suppose $a \sqcup a^c \neq \top$. The prime ideal separation theorem (mathlib's DistribLattice.prime_ideal_of_disjoint_filter_ideal) yields a prime ideal J containing $a \sqcup a^c$ but not $\top$. The indicator of J 's complement,

$$v(x) = \begin{cases} 0 & x \in J \\ 1 & x \notin J, \end{cases}$$

is a valuation. Modularity in the only nontrivial case ($x, y \notin J$) is *exactly* primeness, $x \sqcap y \in J \Rightarrow x \in J \vee y \in J$. And this valuation assigns $v(a) + v(a^c) = 0 \neq 1$. So the complement rule is not one convention among many. Quantified over all valuations, it is a statement *about the algebra* that is equivalent to excluded middle.

Sharp valuations are the points of the locale. The construction above is not ad hoc. It is one half of an exact correspondence. Call v *sharp* if it is $\{0, 1\}$ -valued, a state of full certainty. Sharpness characterizes exactly the prime-ideal indicators:

Theorem 6 (sharp_iff_point). *For every valuation v on a frame Ω , v is sharp iff v is the complement-indicator of a prime ideal.*

This theorem is witnessed by two mutually inverse maps: $v \mapsto Z(v) = \{x \in \Omega : v(x) = 0\}$, which computes the zero-set of a sharp valuation, and the map taking J to its complement-indicator (the valuation of Theorem 5's construction). $Z(v)$ is prime exactly because v is sharp, and the complement-indicator of $Z(v)$ recovers v itself. In locale-theoretic terms, sharp valuations recover the *points* of the locale. A prime ideal is *completely prime* when it is also closed under arbitrary (not just finite) suprema of its members. The finitely prime points of sharp_iff_point sharpen to the completely prime points exactly where Scott continuity is added, recovering points of the locale in the sense used by [Coquand and Spitters 2009; Vickers 2008]:

Theorem 7 (sharp_scott_iff_completelyPrimePoint).
For every valuation v on a frame Ω , v is sharp and Scott-continuous iff v is the complement-indicator of a completely prime ideal.

The hinge and the two correspondences together answer the question “where does classicality live”: whenever excluded middle fails at some a , the failure is witnessed by a point of the locale.

4 Representation: Points, Masses, Mixtures

Every valuation on a finite frame is a probability mass function on points. On infinite frames, this can fail, and Scott continuity marks the boundary between where it still holds and where it does not.

4.1 Finite frames: every valuation is a PMF on points

Let P be the finite poset of sup-irreducible elements of a finite frame. Then, by Birkhoff duality (OrderIso.lowerSetSupIrred), every finite frame is the lattice of *lower sets* of P . We prove the finite representation theorem below on LowerSet P (Valuation.exists_pmf_presentation). Transporting it across the isomorphism above then gives the theorem for every finite frame, not just lower sets. Let us define the *mass* of a point as the discrete derivative of v :

```
def Valuation.mass (v : Valuation (
  LowerSet P)) (p : P) :
  ℝ≥0∞ := v (LowerSet.Iic p) - v (
    LowerSet.Iio p)
```

From the mass at every point, we can fully recover v . The masses also sum to 1, forming a probability distribution on P :

Theorem 8 (finite representation, eq_sum_mass, sum_mass).
For a finite P , any valuation v on Low(P), and any $U \in \text{Low}(P)$, $v(U) = \sum_{p \in U} \text{mass}(p)$, and $\sum_p \text{mass}(p) = 1$.

The proof works by induction on $|U|$: repeatedly remove a maximal point p from U , and use modularity applied to $(U \setminus \uparrow p, \downarrow p)$ to show each removal changes v by exactly $\text{mass}(p)$. Valuation.toPMF gives mass the type PMF on P , i.e. a classical probability mass function. And since lower sets are exactly the opens of the Alexandrov topology, where interior is the identity, the theorem also reads $v = \mu \circ \sqcap$, i.e. every valuation on a finite frame is the classical measure μ applied after taking the interior. In the finite case, this is the converse of the measure-theoretic bridge of Section 7.

The theorem above represented v as a sum of point masses. The dual view represents v as a mixture of sharp *point-valuations*. For each $p \in P$, let δ_p be the valuation with $\delta_p(U) = 1$ if $p \in U$, and $\delta_p(U) = 0$ otherwise:

Theorem 9 (mixture characterization, eq_mix_deltaPoint).
On a finite frame, every valuation v satisfies $v = \sum_p \text{mass}(p) \cdot \delta_p$.

Together with the fact that any mixture of valuations is again a valuation (Valuation.mix), this theorem characterizes the valuations on a finite frame exactly: they are the finite convex mixtures of point-valuations, closed under mixing because modularity is linear. Section 6 shows modularity cannot instead be *derived* from more primitive axioms. It must be posited outright, and doing so is exactly what makes every valuation a mixture of sharp, all-or-nothing certainties.

4.2 The infinite boundary: enter Scott continuity

Both halves of the finite theorem fail for infinite frames:

Theorem 10 (obstruction, exists_valuation_not_point_representable). On Low($\mathbb{N}$), the indicator v of $\top$ is a valuation with $\text{mass}(p) = 0$ for every point p , and $v(\top) = 1$.

No point carries any of the mass. The failure is one of approximation: $v(\top)$ is not the supremum of $v(\text{Iic } n)$ over the finite stages. Requiring that approximation, i.e. Scott continuity, restores the representation:

Theorem 11 (recovery, eq_tsum_mass_of_scott, generalized by isPurelyAtomic_of_scott). Let v be a valuation on Low($\mathbb{N}$). If $v(\top) = \bigsqcup_n v(\text{Iic } n)$ then $v(\top) = \sum'_n \text{mass}(n)$.

The counterexample of Theorem 10 demonstrates a failure of Scott continuity (topIndicator_not_scott). In general, on any poset all of whose principal downsets are finite, Scott continuity implies zero diffuse part without additional preconditions. However, subtraction in $[0, \infty]$ is truncated at 0, so if $\sum'_p \text{mass}(p)$ were greater than $v(\top)$, the expression $v(\top) - \sum'_p \text{mass}(p)$ would still evaluate to 0. Theorem 12 rules out $\sum'_p \text{mass}(p) > v(\top)$ unconditionally, so this subtraction is never actually truncated and its value is the actual result of the subtraction:

Theorem 12 (general decomposition, tsum_mass_le).
For every valuation v on Low(P) and an arbitrary poset P , $\sum'_p \text{mass}(p) \leq v(\top)$.

Rearranging Theorem 12, licensed by the truncation point above, gives $v(\top) - \sum'_p \text{mass}(p) \geq 0$. Every valuation thus splits unconditionally into an *atomic* part carried by points ($\sum'_p \text{mass}(p)$, which is always a sub-probability), and a *diffuse* (nonnegative) remainder $v(\top) - \sum'_p \text{mass}(p)$. This is a constructive Lebesgue-style decomposition. The finite and Scott cases are the purely atomic extremes, and the $\top$ -indicator is the purely diffuse one. The proof of Theorem 12 re-runs the maximal-point induction on each finite subset of P , bounding its partial sum by $v(\top)$, then takes the supremum over all finite subsets to bound $\sum'_p \text{mass}(p)$ itself. Its proof does not directly invoke an excluded-middle case split. However, it relies on the (classical) decidability of the ordering of the

reals (a default in mathlib). Representing the diffuse part on non-spatial frames remains the open frontier. The obstruction there is again that the frame has *too few points*, the same phenomenon behind the undecidability of Section 7.

5 The Dempster–Shafer Reading, Completed

Sections 2–4 independently recover the *shape* of Dempster–Shafer (DS) theory [Dempster 1967; Shafer 1976], including sub-additive complements, a belief/plausibility interval, and mass functions. This section makes the identification exact in both statics and dynamics. In each case, it locates the precise point where excluded middle makes a difference.

5.1 Statics: total monotonicity via constructive inclusion–exclusion

The $\neg\neg$ -stable (*regular*) elements of a frame form a Boolean algebra with meet $\sqcap$ and join $(a \sqcup b)^{cc}$ (Glivenko). Let us call the resulting algebra the *Booleanization* of this set. A DS *belief function* is a capacity on a Boolean algebra that is n -monotone for every n : $\text{Bel}(\bigvee_i A_i) \geq \sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} \text{Bel}(\bigwedge_{i \in T} A_i)$. We show v , restricted to its Booleanization, is such a capacity for every n . The 2-monotone case is immediate from modularity (two_monotone). The full tower follows from the following result:

Theorem 13 (inclusion–exclusion, inclusion_exclusion). *For every valuation on every frame Ω , every finite family $a : \iota \rightarrow \Omega$, and every finite $s \subseteq \iota$, inclusion–exclusion holds with equality for the frame join:*

$$v\left(\bigsqcup_{i \in s} a_i\right) = \sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} v\left(\bigsqcap_{i \in T} a_i\right).$$

We state this in Lean additively, with the odd-cardinality half of the signed sum set against the even-cardinality half, so that no $[0, \infty]$ subtraction occurs. The proof is by induction with modularity and frame distributivity.

Replacing the frame join $\bigsqcup_i a_i$ on the equality’s other side with its double negation $(\bigsqcup_i a_i)^{cc}$, the Booleanization’s own join, yields the following inequality:

Theorem 14 (total monotonicity, infty_monotone). *For every valuation on every frame Ω , every finite family $a : \iota \rightarrow \Omega$, and every finite $s \subseteq \iota$:*

$$\sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} v\left(\bigsqcap_{i \in T} a_i\right) \leq v\left(\left(\bigsqcup_{i \in s} a_i\right)^{cc}\right).$$

The Booleanization’s join is given by $(\cdot)^{cc} \circ \sqcup$, and its meets of regular elements are again regular. Interpreted in this way, Theorem 14 states exactly the ∞ -monotone tower: v restricted to its Booleanization is a belief function.

The proof of Theorem 14 from Theorem 13 follows from the monotonicity along $a \leq a^{cc}$. Here, inclusion–exclusion does *not* fail constructively. It is exact for the intuitionistic

disjunction. Applying the double-negation nucleus (a closure operator preserving finite meets) $(\cdot)^{cc}$ to the join on the equality’s right-hand side can only enlarge it. This follows from the fact that $a \leq a^{cc}$ and v are monotone, which turns the equality into the inequality of Theorem 14. The amount by which the right-hand side is enlarged is exactly the double-negation gap of Section 2.

Additive probability does not become a belief function by weakening an axiom, since modularity for v is never relaxed. What changes is only the join used to state the tower: the frame join $\sqcup$ is replaced by its double-negation closure $(\cdot)^{cc} \circ \sqcup$. A belief function is thus the same additive valuation v , evaluated against that closure rather than against the frame join. With $\text{Pl}(a) = 1 - v(a^c)$, the belief–plausibility interval has width exactly the slack (plausibility_sub_self). The DS “ignorance” is boundary measure, not psychology.

5.2 Dynamics: the conditioning hinge

DS theory carries two update rules that coincide classically. *Geometric* conditioning renormalizes belief, and under the DS dictionary it is the localic posterior $v(a \sqcap b)/v(b)$ of Section 2. *Dempster’s rule of conditioning* [Shafer 1976] discards the mass incompatible with the evidence and renormalizes by its plausibility:

$$v(a \parallel b) = \frac{v(a \sqcup b^c) - v(b^c)}{1 - v(b^c)}.$$

Both are again valuations (dempsterCond). Their modularity is the frame distributivity $(a \sqcup b^c) \sqcap (a' \sqcup b^c) = (a \sqcap a') \sqcup b^c$. Their disagreement is given by the *excluded-middle gap* $\text{emGap}(a, b) = v(a) - v(a \sqcap (b \sqcup b^c))$, i.e. the mass of a outside b ’s instance of excluded middle:

Lemma 15 (sup_compl_decomp). *For every valuation v on every frame Ω and every $a, b \in \Omega$, $v(a \sqcup b^c) = v(b^c) + v(a \sqcap b) + \text{emGap}(a, b)$.*

Dempster’s numerator therefore exceeds Bayes’s by exactly $\text{emGap}(a, b)$. This gap depends on a , but whether it vanishes at every a , in turn, depends on a single number, the slack at the evidence b :

Theorem 16 (the conditioning hinge, dempsterCond_eq_condVal_iff_slack). *For every valuation v on every frame Ω and every $b \in \Omega$ with $v(b) \neq 0$, $(v(a \parallel b) = v(a \sqcap b)/v(b))$ for every $a \in \Omega$ iff $\text{slack}(b) = 0$.*

The hypothesis $\text{slack}(b) = 0$ eliminates the gap $\text{emGap}(a, b)$ for every a and makes the two normalizers coincide, since $1 - v(b^c) = v(b)$. Conversely, testing at $a := b$, geometric conditioning gives certainty 1, while Dempster gives $v(b)/(1 - v(b^c))$, (which is 1 only if the slack vanishes). Theorem 16 is the valuation-relative analogue of Theorem 5: both are iff statements about the same proposition, $b \sqcup b^c = \top$. The static hinge is an equation in the frame itself, between $b \sqcup b^c$ and $\top$, with no valuation

involved. The dynamic hinge tests the same proposition through v : $\text{slack}(b) = 0$ is exactly $v(b \sqcup b^c) = v(\top)$. Dempster and Bayes therefore coincide at b whenever v cannot distinguish $b \sqcup b^c$ from $\top$, even if the two differ in the frame itself. In the measure model, Dempster and Bayes disagree precisely when the evidence's boundary carries mass.

6 The Cox Program, Dissected

Cox's theorem [Cox 1946; Jaynes 2003] derives the probability calculus from desiderata on plausibility. After Halpern's counterexample [Halpern 1999], its repaired forms [Arnborg and Sjödin 2000; Horn 2003] fix the calculus up to monotone regratuation. Every known derivation is classical, and Section 3 showed why: the negation axiom R3 is excluded middle. What survives of the program when R3 is dropped? Mechanization forced a sharper answer than we had conjectured, in three parts. The product rule is logic-free. The sum rule is irreducible. And the naive constructive statement is unsatisfiable.

6.1 A Cox structure without negation

```
structure CoxModel (Ω : Type*) [Order.
  Frame Ω] where
  pl : Ω → Ω → ℝ          -- pl a b :
    plausibility of a given b
  mono_left : ∀ c, Monotone fun a => pl a
    c
  pl_bot : ∀ c, pl ⊥ c = 0
  pl_top : ∀ c, pl ⊤ c = 1
  F : ℝ → ℝ → ℝ          -- conjunction
    functional
  conj : ∀ a b c, pl (a ⊓ b) c = F (pl a (
    b ⊓ c)) (pl b c)
  F_assoc : ∀ x y z, F (F x y) z = F x (F
    y z)
  F_strictMono_left : ∀ y, 0 < y →
    StrictMono fun x => F x y
```

The deliberate omission is any field relating $\text{pl}(a^c, b)$ to $\text{pl}(a, b)$. That missing field is exactly R3. Two details are themselves checker-forced. Strict monotonicity of F must be restricted to $0 < y$, because the boundary axioms force $F(x, 0) = 0$ for all x , so the unrestricted axiom makes the structure *uninhabited* and every theorem about it vacuous. We guard against this permanently: `coxModelENNReal` is an explicit model on the non-Boolean chain $[0, \infty]$, with no appeal to excluded middle at the object level. Deciding $a = \top$ there is a constructor check, and the formalization supplies the computable Decidable instance witnessing this (`decEqTopENNReal`), so the axiom class is inhabited by a genuinely non-classical instance.

6.2 The sum rule is irreducible

Classically, additivity is *derived*: R3 turns disjunction into negated conjunction via De Morgan, and the product rule does the rest. Constructively that route is closed, since $a \vee b \neq \neg(\neg a \wedge \neg b)$ in general. And this is not an artifact of one proof strategy:

Theorem 17 (`modularity_irreducible, no_disjunction_functional`). *There is a five-element frame (the lower sets of the poset $o < a, o < b$ with a, b incomparable) and a monotone, normalized assignment q on it that is additive on disjoint joins yet not modular. Moreover no binary operation S on values satisfies $q(x \sqcup y) = S(q(x), q(y))$ for all x, y . The pairs $(\downarrow a, \downarrow a)$ and $(\downarrow a, \downarrow b)$ have identical value-pairs $(\frac{1}{2}, \frac{1}{2})$ but joins valued $\frac{1}{2}$ and 1.*

A disjunction's plausibility is not a function of its disjuncts' plausibilities, unlike a conjunction's, which the `conj` field reduces to values via conditioning. So a constructive Cox theorem *must* posit the sum-rule half. Modularity is the posit, and Theorem 9 is its independent justification: modularity says exactly that the calculus is closed under mixing certainities.

6.3 The theorem, corrected and proved

Our original target statement asked for a strictly monotone regratuation $g : \mathbb{R} \rightarrow [0, \infty]$ with $g(0) = 0$. The checker refused, for a reason no referee had caught on paper: strict monotonicity forces $g(-1) < g(0) = 0$, impossible in $[0, \infty]$. The statement was not too hard. It was *unsatisfiable*, and the repair (strictness only on $[0, 1]$, where plausibilities live) is exactly the domain hygiene that Halpern's critique of the classical theorem turns on. The corrected statement is a theorem:

Theorem 18 (`constructive_cox`). *For every frame Ω , every Cox model on Ω whose unconditional plausibility is modular regratuates to a valuation: there exist $v : \text{Valuation } \Omega$ and g strictly monotone on $[0, 1]$ with $g(0) = 0$, $g(1) = 1$, and $v(a) = g(\text{pl}(a, \top))$ for every $a \in \Omega$.*

The regratuation for the sum-rule half is undemanding ($g = \text{ENNReal.ofReal}$). The substance lives in the product-rule half, showing F regratuates to multiplication, which is Aczél's associativity theorem [Aczél 1966], a statement of pure analysis whose type never mentions Ω . The constructive/classical divide is therefore *entirely* in the sum rule. `mathlib` has no Aczél theorem, no Hölder-style embedding for ordered semigroups on intervals, and no Cauchy functional equation under monotonicity (only under continuity), so we build the analytic core from scratch ($\approx 1,100$ lines): a bundled `Scale` of Aczél hypotheses, a generator defined as the dyadic limit $g(b) = \bigsqcup_n (\text{count of } 2^n\text{-th roots under } b) / 2^n$, and the Hölder-style conclusion, a normalized, strictly monotone

additive generator and hence a strictly monotone multiplicative one, on the positive cone (`exists_mul_generator`, `aczelStatement_cone`). The Archimedean property is *derived* from continuity and positivity rather than assumed, and the interval form of divisibility is likewise a consequence of the intermediate value theorem (`exists_diag_eq`), though the bundled `Scale` takes divisibility as a hypothesis. Appendix A gives the construction. The residual analytic gap, continuity of the generator and extension below the unit, is stated honestly there and in Section 9, and is witnessed non-vacuously by the log-sum-exp archetype, for which the global continuous generator is exp.

7 Two Grounding Models

Two concrete models anchor the abstract theory. One shows every classical measure *contains* an intuitionistic probability. The other shows the intuitionistic theory cannot collapse back without solving the halting problem.

7.1 Measures through the interior operator

Gödel–McKinsey–Tarski interpret intuitionistic logic in modal S4, semantically: propositions are *open* sets, and $\Box$ is topological interior [Gödel 1933; McKinsey and Tarski 1944]. Our valuations are the quantitative face of that translation:

Theorem 19 (`Measure.toValuationOpens`). *Let μ be a probability measure on a space X . Then $U \mapsto \mu(U)$ on the frame $O(X)$ is a valuation. Equivalently, $v = \mu \circ \text{int}$ as a function of arbitrary sets, that is, $v(A) = \mu(\Box A)$.*

Modularity is $\mu(U \cup V) + \mu(U \cap V) = \mu(U) + \mu(V)$. The slack of an open U is then $1 - \mu(U) - \mu(\text{int } U^C)$, which is the measure of the boundary ∂U . *Slack is boundary mass*, and now provably so in general:

Theorem 20.

(`Measure.toValuationOpens_slack_eq_frontier`)
For any probability measure μ on a topological space X and any open U , $\text{slack}(U) = \mu(\partial U)$.

The Heyting complement of U in $O(X)$ is the interior of its set complement, so the undecided region, the complement of $U \sqcup U^C$, is exactly $\bar{U} \setminus U = \partial U$. The development also keeps the weaker complement sub-additivity inequality (`interiorMeasure_add_compl_le`), and the Sierpiński model below instantiates the identity in the specific case that isolates the halting problem. The DS “ignorance” of Section 5 is, in this model, the measure charged to the frontier where a finite-precision observer can never decide membership, a structural quantity with no epistemic subject. This also exhibits the modal identification $\text{Bel} = P(\Box \cdot)$ of the DS literature [Ruspini 1986] arising from the logic itself. Theorem 8 is the converse in the finite case, and the general converse is the open frontier of Section 4.

Remark 21 (`isDeMorgan_opens_iff_extremallyDisconnected`). Section 2 noted that imposing weak excluded middle on the whole frame collapses slack to pure double-negation instability. For $\Omega = O(X)$ this global condition is not abstract: it holds exactly when X is *extremally disconnected*, meaning the closure of every open set is already open. These are the Stonean spaces (Stone–Čech remainders, Stone spaces of complete Boolean algebras). As before, this is a parallel development and an obvious complement to the fully constructive theory the paper builds on $O(X)$. No other result here assumes it.

7.2 The computability guard

The design constraint on every axiom set in this paper is that it must not silently force $\text{slack} \equiv 0$. Otherwise the “constructive” theory is Kolmogorov in disguise. Our guard is a model in which collapse is not merely false but *uncomputable*.

A semidecidable proposition is an open in the Sierpiński topology on `Prop`: verifiable by a finite computation, never refutable by one. On the three-open frame $O(\mathbb{S}) = \{\perp, \text{halts}, \top\}$ we prove the Σ_1 asymmetry as a theorem, $\text{halts}^c = \perp$ (every nonempty open of $\mathbb{S}$ contains `True`, so nothing partially refutes a semidecidable event), and build the valuation from a genuine measure $p \cdot \delta_{\text{True}} + (1 - p) \cdot \delta_{\text{False}}$ through the bridge above. The slack computes end to end (`slack_eq_boundary`):

$$\text{slack}(\text{halts}) = 1 - p = \mu(\partial\{q \mid q\}) = \mu\{\text{False}\},$$

the mass of the point where the computation stays silent forever.

The guard itself is grounded in `mathlib`’s computability theory. For a code c and input n , the *sharp classical readout* assigns the slack-free valuation answering “does c halt on n ?” with certainty, weight 1 or 0 by the actual halting fact. The family is classically well defined, but no algorithm can read it off:

Theorem 22 (`sharpReadout_not_computable`). *Deciding whether the sharp classical readout assigns weight 1 to the halting event is exactly the halting problem, hence not computable* (`ComputablePred.halting_problem`).

Any axiom strong enough to force classical collapse therefore demands of its models a state that cannot be computed. Slack is not a defect the constructive theory should apologize for. It is the price of a computable one. This guard plays for the sum rule the role the explicit `CoxModel` instance plays for the product rule (Section 6): each axiom set in the paper ships with a machine-checked witness that it has non-degenerate models.

8 Related Work

Informal precedents. That probability might be paired with intuitionistic logic is an old if scattered thought. Weatherson develops intuitionistic probability axiomatically and

defends it with Dutch-book arguments [Weatherson 2003]. Paris treats non-classical uncertain reasoning broadly [Paris 1994]. The identification $\text{Bel} = P(\Box \cdot)$ connecting belief functions to modal logic is classical [Dempster 1967; Ruspini 1986; Shafer 1976]. On the localic side, valuations on frames are standard objects of constructive measure theory, with Vickers's work and the Coquand–Spitters development as canonical references [Coquand and Spitters 2009; Vickers 2008], normally with Scott continuity built in and with additive, Riesz-flavored goals. What we take from Section 4 is that Scott continuity is a substantive assumption (it decides point-representability on the frames we examine), so our base structure omits it. The Cox line, through Cox, Jaynes, Halpern's counterexample, and the Van Horn and Arnborg–Sjödín repairs [Arnborg and Sjödín 2000; Cox 1946; Halpern 1999; Horn 2003; Jaynes 2003], is classical throughout. Van Horn himself observes that dropping R3 yields a two-number theory, which is exactly the belief/plausibility pair our slack separates. To our knowledge, no prior work, informal or formal, states either hinge theorem: complement rule $\iff$ excluded middle (Theorem 5), or Dempster = Bayes $\iff$ zero slack at the evidence (Theorem 16).

Prior formalizations. Three bodies of mechanized work are adjacent, and none overlaps. ALEA [Audebaud and Paulin-Mohring 2009] formalizes a discrete distribution monad in Coq for randomized programs: additive probability, no event logic in our sense. Recent work in cubical Agda formalizes synthetic probability via Markov categories [Sargsyan et al. 2023], where again the objects are additive distributions. Closest is the HoTT/Coq line of Faissole and Spitters on *valuations* over lower reals [Faissole and Spitters 2017], done constructively with Scott continuity and aimed at integration (Riesz) and a Giry-style valuation *monad*. Their valuations are the additive, continuous cousins of ours, and their monad is the Scott-continuous instance of the question examined in Section 9. The delta of this paper is not the word “valuation” but the development on top of it: the non-additive complement regime, slack and its decomposition, both hinge theorems, the DS bridge with total monotonicity and the two conditionings, the representation/mixture/Scott-atomic suite, the corrected Cox theorem with its irreducibility counterexamples, the Aczél construction, product rigidity, and the computability guard. We claim first mechanization for that development, not for valuations on locales as such.

Non-Boolean probability elsewhere. Quantum probability on orthomodular lattices, with Gleason's theorem [Birkhoff and von Neumann 1936; Gleason 1957], is the long-standing precedent for re-basing probability on a non-Boolean event logic. Free probability, fuzzy measures, and imprecise probability [Walley 1991] vary other parameters (independence, additivity, single-valuedness). Wolpert and Kinney model mathematics itself as a stochastic

process, replacing binary theoremhood with a probability distribution [Wolpert and Kinney 2024]. Theoremhood is a semidecidable event, so the computability guard of Section 7 bears directly on that program: slack-free credences about such events are uncomputable. Our variation is specifically the *logic*, holding the Cox-style methodology fixed. Within DS theory, the two conditionings and their inequivalence are classical [Fagin and Halpern 1991; Shafer 1976]. What is new in Theorem 16 is the exact logical characterization of their agreement, and its mechanization.

9 Conclusion: Probability with the Logic as a Parameter

We set out from the observation that Kolmogorov's axioms fix a logic before they fix an arithmetic, and that nothing in the concept of graded belief privileges the Boolean choice. The machine-checked results bear the framing out with a precision informal treatments have not reached. The entire classical surplus is one axiom, whose statical face is the complement rule (Theorem 5) and whose dynamical face is the agreement of Dempster and Bayesian conditioning (Theorem 16). Everything else survives the intuitionistic substrate intact: conditioning, Bayes symmetry, disjoint additivity, total probability over genuine partitions, and inclusion–exclusion itself (Theorem 13). What the weaker logic adds is structure, not damage: slack with its two-gap decomposition, the atomic/diffuse divide, the belief-function reading through $(\cdot)^{\text{cc}}$, and a computability-theoretic reason the additions are not optional.

Open problems. Three are mathematical. First, representing the diffuse part on non-spatial frames (the localic Riesz direction). Second, closing the analytic residue of the Aczél development, namely continuity of the generator and the off-cone extension (the cone results and the log-sum-exp archetype are done). Third, a genuine uniqueness theorem: deriving modularity from a more primitive desideratum, knowing (Theorem 17) that it cannot come from disjunction data alone, with the mixture characterization (Theorem 9) as the current best justification. A fourth is foundational hygiene at scale: a machine-checked Dutch-book theorem in the style of [Weatherson 2003], which to our knowledge no proof assistant has for *any* logic.

The functor is literal, the monad is delimited. Within a fixed doctrine, that is, a fixed choice of logic, the connection to the categorical picture is concrete. Pushforward along frame homomorphisms preserves all four valuation axioms, making $\text{Val}(-)$ a functor on locales, the localic face of the functor part of the Giry monad [Giry 1982]. This is now mechanized end to end (Valuation.comap builds the pushforward, and $\text{comap_id/comap_comp}$ check the two functor laws against identity and composition of frame homomorphisms). The point valuations δ_p of Section 4 are the

evident unit (these *are* formalized), and Theorem 8 shows that on finite frames Val agrees, on objects, with the classical finite-distribution monad on points. The *multiplication* is where the substance is. Flattening a valuation of valuations is integration, hence product valuations and Fubini, and by Kock’s identification of commutative monads with Fubini theorems [Kock 2012], whether statistics-as-kernel-composition transports to a weaker logic is precisely the question whether that logic’s valuation functor is a commutative monad. Two machine-checked results now delimit the multiplication from both sides.

On the chain product the question of *which* product valuation to take does not arise. The frame $\text{Low}(\mathbb{N} \times \mathbb{N})$ is *rigid*. Every lower set is a *finite* union of rectangles: beyond a stabilization height every occupied column is full, so the full columns and finitely many rows account for everything (exists_rect_decomposition). The rigidity mechanism is general: by the inclusion–exclusion equality, two valuations agreeing on any $\sqcap$ -closed family agree on all finite joins of its members (eq_on_finsetSup_of_eq_on), and the chain product is the instance where that family is the rectangles. So the product valuation there, whenever it exists, is unique, with no continuity assumed (Valuation.product_unique). We had expected the opposite, diffuse mass free to choose among the infinities of the product, and our candidate counterexamples all died on the same rock: lower sets of chain products detect only finitely much of the boundary.

They were right in spirit and wrong in venue. On the powerset frame $\text{Set}(\mathbb{N} \times \mathbb{N})$, whose elements are far from finite unions of rectangles, non-uniqueness is a theorem. The indicator of an ultrafilter is a valuation (ultrafilterValuation), and any ultrafilter on a product splits rectangles through its marginal pushforwards (rect_mem_iff). Taking a free ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the two iteration orders of its tensor with itself and its diagonal pushforward all carry identical values on every rectangle, yet they differ on the upper triangle $\{(m, m') \mid m < m'\}$ (product_valuation_not_unique). In particular the two orders of iterated integration of the same marginal data disagree: *Fubini fails*, concretely (fubini_fails_for_valuations). Compare the Fubini pathologies of non-additive integration [Denneberg 1994; Ghirardato 1997]. By Kock’s correspondence, no canonical commutative multiplication exists for unrestricted valuations. On the powerset frame these valuations are finitely additive probability charges in the sense of [Rao and Rao 1983], so this half of the dichotomy is not about intuitionism at all: the freedom lives already in the classical finitely additive world, and dropping continuity is what lets it in. Anyone who wants a monad must therefore restrict, and the dichotomy says along which axis: the frames, or the valuations.

The hard problem. The question *upstream* of the multiplication deserves to have its hardness named rather than implied, and what follows is our assessment rather than a theorem. A monad needs a category for Val to be an endofunctor on. The space of valuations must itself be an object, a “space” whose opens one can speak of. For Scott-continuous valuations that space is a locale and monad structure is available, constructively, through the probabilistic powerdomain line [Coquand and Spitters 2009; Jones and Plotkin 1989; Vickers 2008]. Yet *even there*, exhibiting a cartesian closed category closed under the construction is the Jung–Tix problem, open since 1998 [Jung and Tix 1998]. Dropping continuity makes the difficulty prior rather than merely worse: the standard presentations of the valuation space use continuity essentially. One route to a well-posed question does exist. The localic treatments present the space of continuous valuations as the classifying locale of a propositional geometric theory [Coquand and Spitters 2009; Vickers 1989], with generators asserting $q < v(a)$ for rational q and axioms for monotonicity, normalization, modularity (expressible with rational witnesses and countable disjunctions), and continuity. Deleting the continuity axiom leaves a geometric theory, so a classifying locale still exists on general grounds, and classically its points are exactly the valuations of this paper. We have verified none of the structure this candidate would need. The unit looks unproblematic, but the multiplication requires integrating the evaluation maps against a merely modular, discontinuous valuation, and there the sum of two lower-real-valued maps is an infinite join, exactly where the diffuse part of Theorem 10 leaks. We offer the formulation as a target, not a result.

The strategy with the best modern track record is to carve a fragment rather than to conquer the whole. On the measure-theoretic side, the failure of cartesian closure for measurable spaces was answered by quasi-Borel spaces [Heunen et al. 2017]. On the domain-theoretic side, commutativity of the full valuations monad on DCPO is open, and the response is a hierarchy of commutative *submonads* generated from simple valuations [Jia et al. 2021]. Our decomposition theorem names the corresponding fragment here: the atomic part. As a first machine-checked step in that direction we prove that the countably-presented fragment composes correctly with no space structure needed at all. A countable mixture of valuations is a valuation (mixCountable), a mixture of mixtures flattens to a mixture with product weights, which is discrete Chapman–Kolmogorov (mixCountable_mixCountable), and the one-point mixture is the identity (mixCountable_unique), all by the unconditional Fubini of $[0, \infty]$ sums over sigma types. And in exact contrast with the failure above, the fragment is *commutative*: iterated countable mixing is order-independent (mixCountable_comm). Commutative exactly where countable. This is associative, unital, commutative kernel composition for countably-atomic data. It is not yet

a monad: the presentations are not canonical, we do not quotient by “presents the same valuation,” and the diffuse part is exactly what escapes every countable presentation. The ledger, then. The unit, the finite level, countable kernel composition, and its commutativity are proved. Rigidity on chain products and non-uniqueness on the powerset frame are proved, so the general multiplication is settled: it is not canonically determined. What remains is the ambient category, a formulation problem before it is anything else.

Future work on the monad. Two continuations are concrete. First, quotient the countable presentations by “presents the same valuation” and package the result as a Kleisli category of countably-atomic kernels, checking that mixing descends to the quotient. That would upgrade composition-on-data to an honest categorical object, and we expect it to be a development of its own. Second, mechanize the classifying-locale formulation sketched above, which requires infrastructure mathlib currently lacks entirely: classifying locales of propositional geometric theories. Both continuations inherit the boundary drawn here. Whatever structure exists must live on a fragment, because the unrestricted multiplication is now provably not canonical.

Across doctrines, by contrast, we make no functoriality claim. There is no ambient category in which “logic $\mapsto$ its probability theory” is a functor, and each column requires re-deriving what a state *is* from the desiderata. What is uniform across columns is the *method*, and this paper’s execution of it is the Heyting column.

The program. If the logic is a parameter, this paper computed one new value of it. The table wants filling in. Linear logic is the most inviting column for this community. Its event algebras are quantales, its conjunction is resource-sensitive, and a “valuation” there should grade the *consumption* of evidence. The analogue of Gleason’s theorem or of our representation theorems (which states are compatible with the algebra?) is wide open, as is the right analogue of slack for the failure of contraction. The probabilistic semantics of linear logic that already exist, probabilistic coherence spaces [Danos and Ehrhard 2011], start from the additive side. A Cox-style derivation over a quantale would start from the logic. The methodology of this paper, to state the classical desiderata, delete the axioms the logic no longer licenses, and let a proof assistant referee what survives, what splits, and what becomes unsatisfiable, is, we suggest, the right instrument for that program. Each cell of the table is exactly the kind of delicate, counterexample-rich boundary work that mechanization is for.

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A The Aczél Development

This appendix sketches the from-scratch construction behind the product-rule half of Theorem 18. The body stands without it. The target shape is Aczél’s associativity theorem [Aczél 1966]: a suitably regular associative $F : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ reg graduates to multiplication, $G(F x y) = G(x) \cdot G(y)$ for all x, y and a strictly monotone G . mathlib offers no route. It has no associativity representation, no Hölder-style embedding of ordered semigroups on intervals, and no Cauchy functional equation under monotonicity (only under continuity).

Hypotheses. A Scale bundles: associativity of F , divisibility $\forall c \exists x, F x x = c$, positivity on the cone $x < F x c$, joint order-preservation, joint continuity, a unit u , and identity at $-\infty$, meaning $F x z \rightarrow x$ as $z \rightarrow -\infty$. Order-preservation was absent from our first bundle, and the additivity proof needs it. Divisibility is taken as a hypothesis in the bundle, although its interval form is a consequence of the intermediate value theorem (exists_diag_eq). Non-vacuity: *log-sum-exp*, $F x y = \log(e^x + e^y)$ with $u = 0$, satisfies every field (logSumExpScale), and for it the intended generator is globally exp.

Construction. Write $a^{(n)}$ for n right-associated copies of a under F (Fpow). Associativity alone makes $n \mapsto a^{(n)}$ a semigroup morphism from $(\mathbb{N}, +)$, with no commutativity needed. Commutativity is a *corollary* of the generator, not a prerequisite. One fact usually assumed is here *derived*: unboundedness of the iterates. Were they bounded, monotone convergence would give a fixed point $FL a = L$, contradicting positivity, and unboundedness yields the Archimedean property. Divisibility supplies a tower of 2^n -th roots $u \succ u^{1/2} \succ u^{1/4} \succ \dots$ (droot) with the coherence law that 2^f fine copies recombine one coarse mark. Counting how many copies of the n -th root fit under b (dcount,

well defined by the Archimedean property) gives dyadic approximants $g_n(b) = \text{dcount}(b, n)/2^n$. The refinement sandwich $2N + 1 \leq N' \leq 2N + 2$, where N and N' are the counts at consecutive refinement levels n and $n + 1$, makes them monotone and Cauchy, and the generator is their limit, $g(b) = \bigsqcup_n g_n(b)$.

Results (all sorry-free, standard axioms). On the cone $\{x \mid u \leq x\}$ we prove three things. Normalization: $g(u) = 1$. Additivity: $g(F x y) = g(x) + g(y)$, by squeezing super- and sub-additive counting bounds ($\text{dcount } x + \text{dcount } y + 1 \leq \text{dcount}(F x y) \leq \text{dcount } x + \text{dcount } y + 3$) against a vanishing 2^{-n} error term. And *strict* monotonicity, proved without any continuity of g by a count-amplification argument: a shrinking root inserted between $x < y$ contributes a full extra block of 2^{n-M} copies, M being the refinement level of the inserted root. Exponentiating gives the multiplicative form, a strictly monotone G with $G(F x y) = G(x) G(y)$ on the cone (exists_mul_generator), which is the conclusion shape of Aczél’s theorem restricted to the cone (aczélStatement_cone). A bounded reorientation $\bar{G} = \exp(-g)$ transports it to the $[0, 1]$ -valued conjunctive orientation of Section 6.

Honest residue. The constructed g is discontinuous at the unit (it is 0 below u), so matching the classical statement verbatim still needs the off-cone extension (a group completion) and continuity of the extension. That is genuine analysis, open here, but *witnessed*: for the log-sum-exp archetype the global continuous generator exists (hasOrderedGenerator_logSumExp). Of Hölder’s classical hypotheses, the Archimedean property is a theorem in this development rather than an assumption, and the interval form of divisibility follows from the intermediate value theorem. Only the embedding’s global regularity remains.